18 · Case 14, part 1: rods, cones, and red-green colour blindness

The words of this track, as it says them. Bold brown is the examiner.

Exam case 14. Red-green colour blindness.

Simon is colour blind, which affects his ability to tell colours apart, most of all red and green. He wants to understand why he has it, when neither of his parents do. He's also heard that it's more common in males, and is wondering if he'll pass it on to any sons he might have.

Two tracks. Her recorded session talks this whole case through, and keys her quiz ten out of ten on screen, so these answers are hers.

Seven marks. What is red-green colour blindness? Distinguish between cones and rods, and where they're found in the eye. Distinguish their functions, using photopic, scotopic, colour, central and peripheral vision. And define red-green colour blindness in terms of the inability to observe certain light frequencies.

The hook: cones see colour, so red-green colour blindness is faulty red and green cones.

Cones are for colour vision: central vision, in good light, and much more detailed.

Rods are for non-colour vision: black, white, and the grey scale. They're more sensitive to light, so they work even in dim light.

Where they are: the macula has both cones and rods. The fovea has only cones.

Red-green colour blindness is a genetic defect, an anomaly rather than a disease, that causes dysfunction of the cones of the retina. It affects the L cones, for long wavelength, red light, and the M cones, for medium wavelength, green light. So red and green are hard to tell apart. It can be detected with a colour-blindness test.

And in the question's own words: cones are photopic, for bright light, colour and central vision. Rods are scotopic, for dim light, and peripheral vision. She says good light and dim light; photopic and scotopic are the question's names for the same idea.

Once more. Rods, cones, and colour blindness.

Cones: colour, central vision, in good light, with more detail: photopic. Rods: the grey scale, more sensitive to light, working in dim light and for peripheral vision: scotopic. The macula has both; the fovea has only cones. Red-green colour blindness is a genetic defect of the retina's cones: faulty L cones, for long wavelength red light, and M cones, for medium wavelength green, so red and green are hard to tell apart. A colour-blindness test detects it.